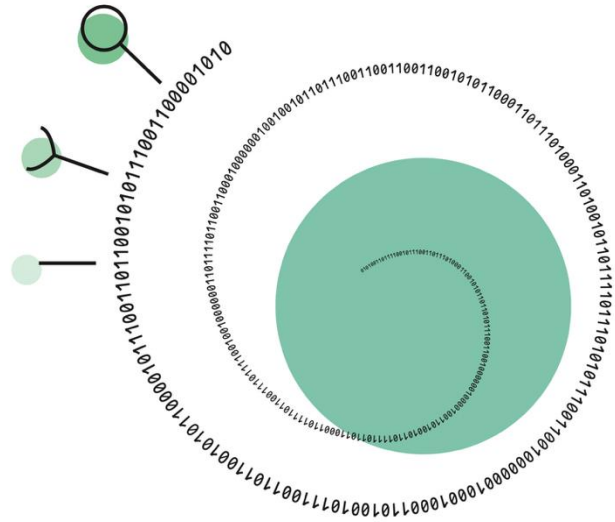




Single Cell Data Analysis Course

Cluster-based cell type annotation

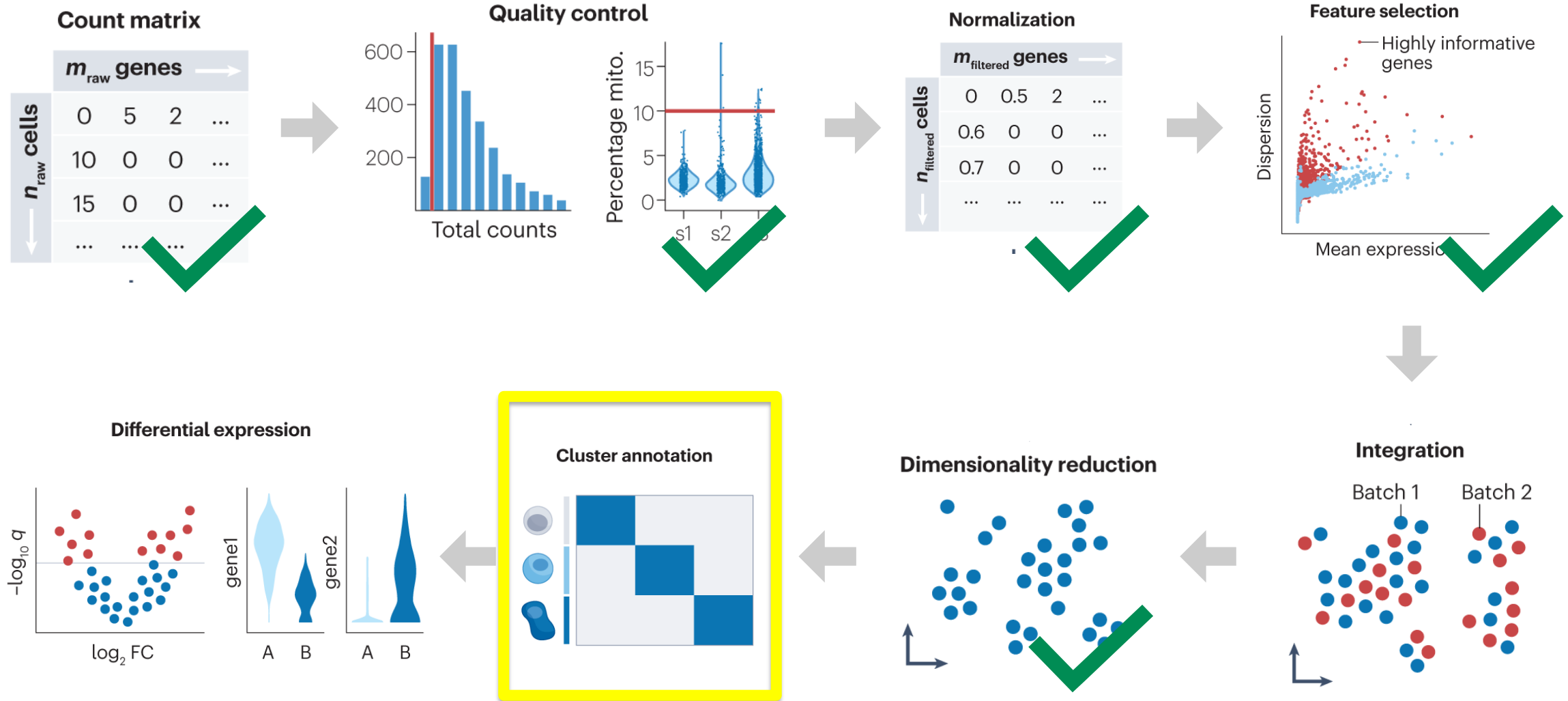
Lisa Buchauer

Professor of Systems Biology of Infectious Diseases

Department of Infectious Diseases and Intensive Care

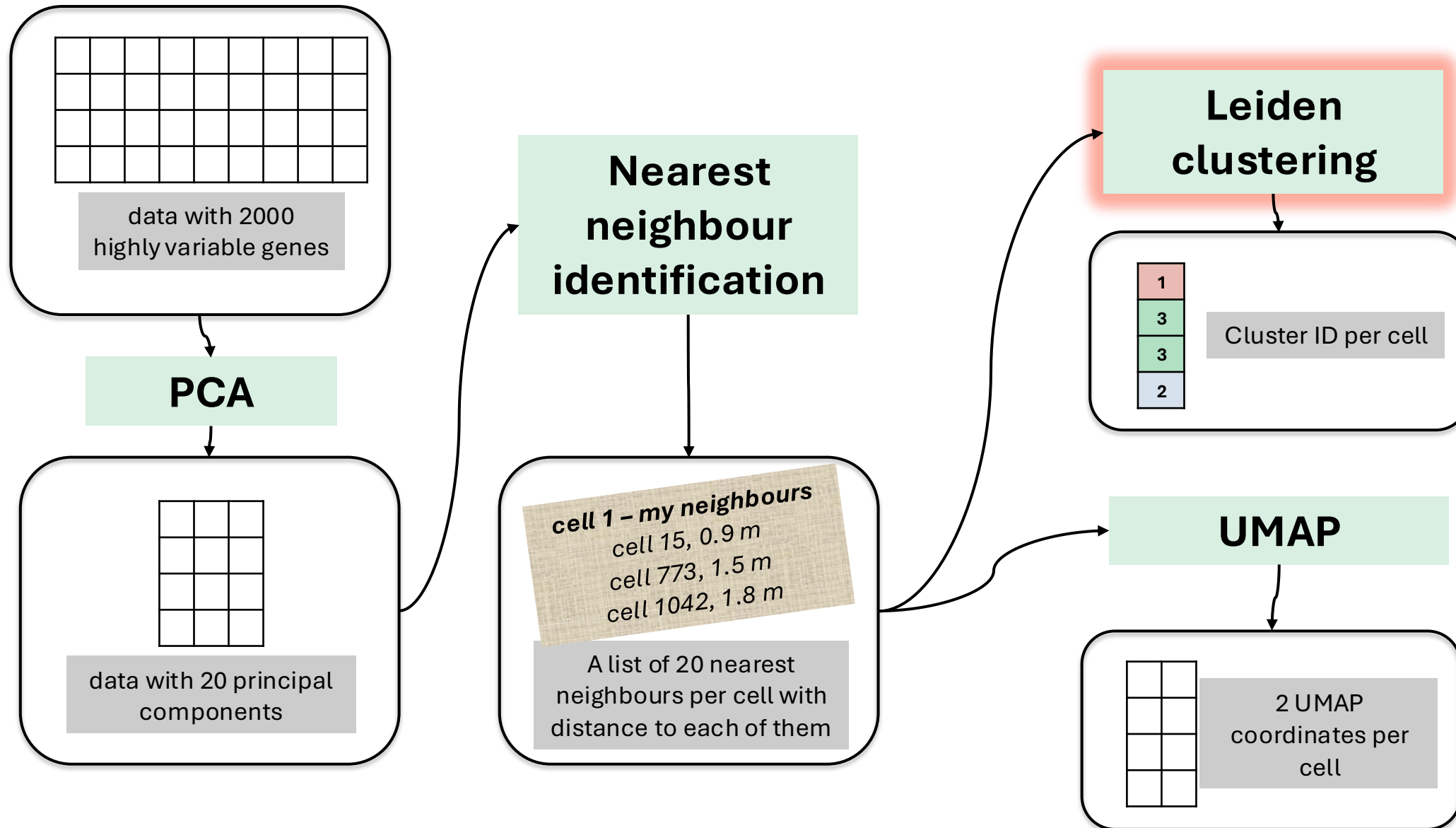
Charité - Universitätsmedizin Berlin

Today

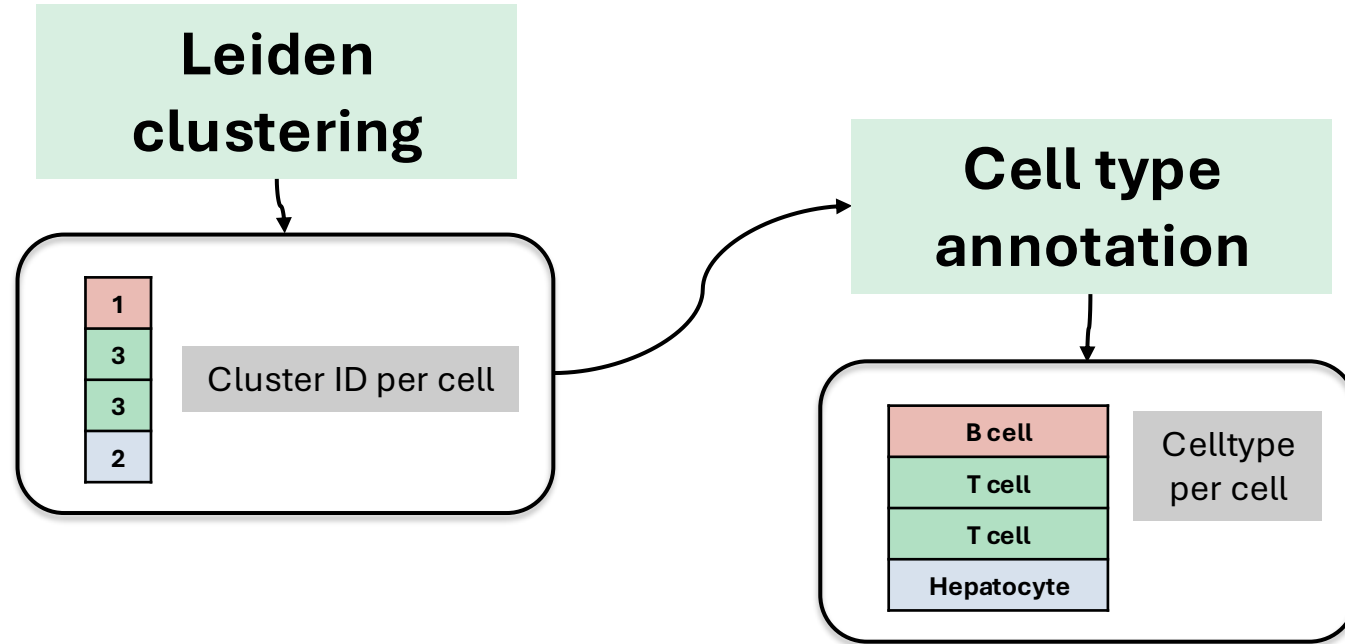


Heumos, L., Schaar, A.C., Lance, C. et al. Best practices for single-cell analysis across modalities. Nat Rev Genet 24, 550–572 (2023). <https://doi.org/10.1038/s41576-023-00586-w>

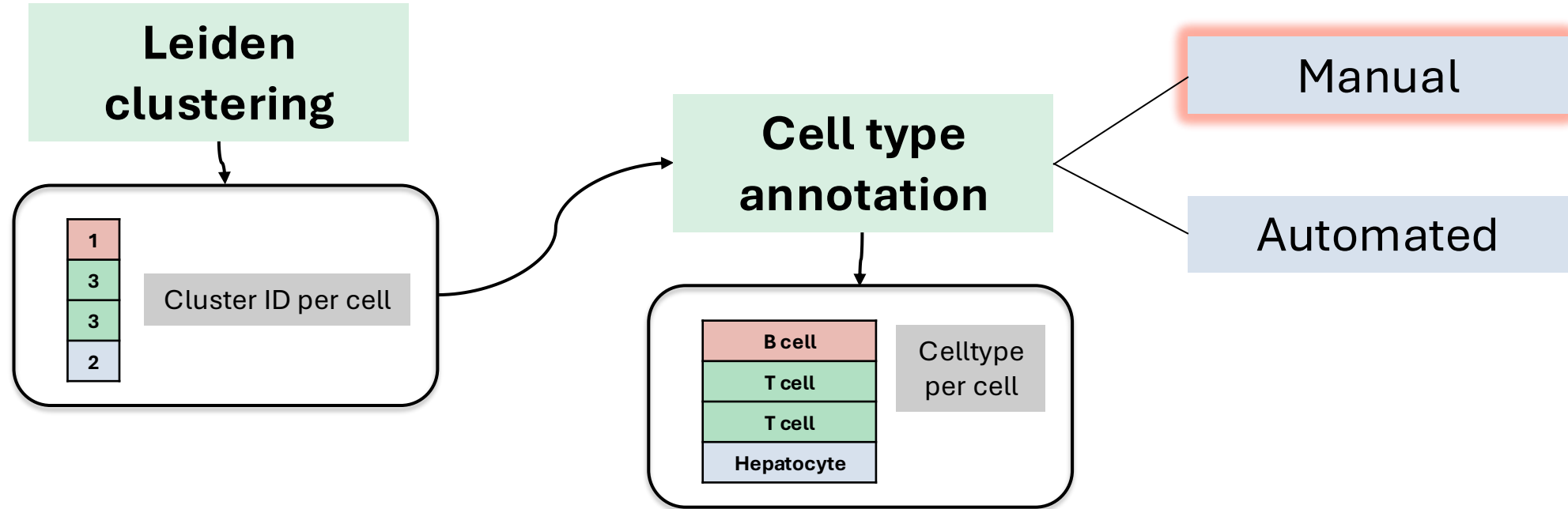
Recap: Where we stand after a whole lot of processing



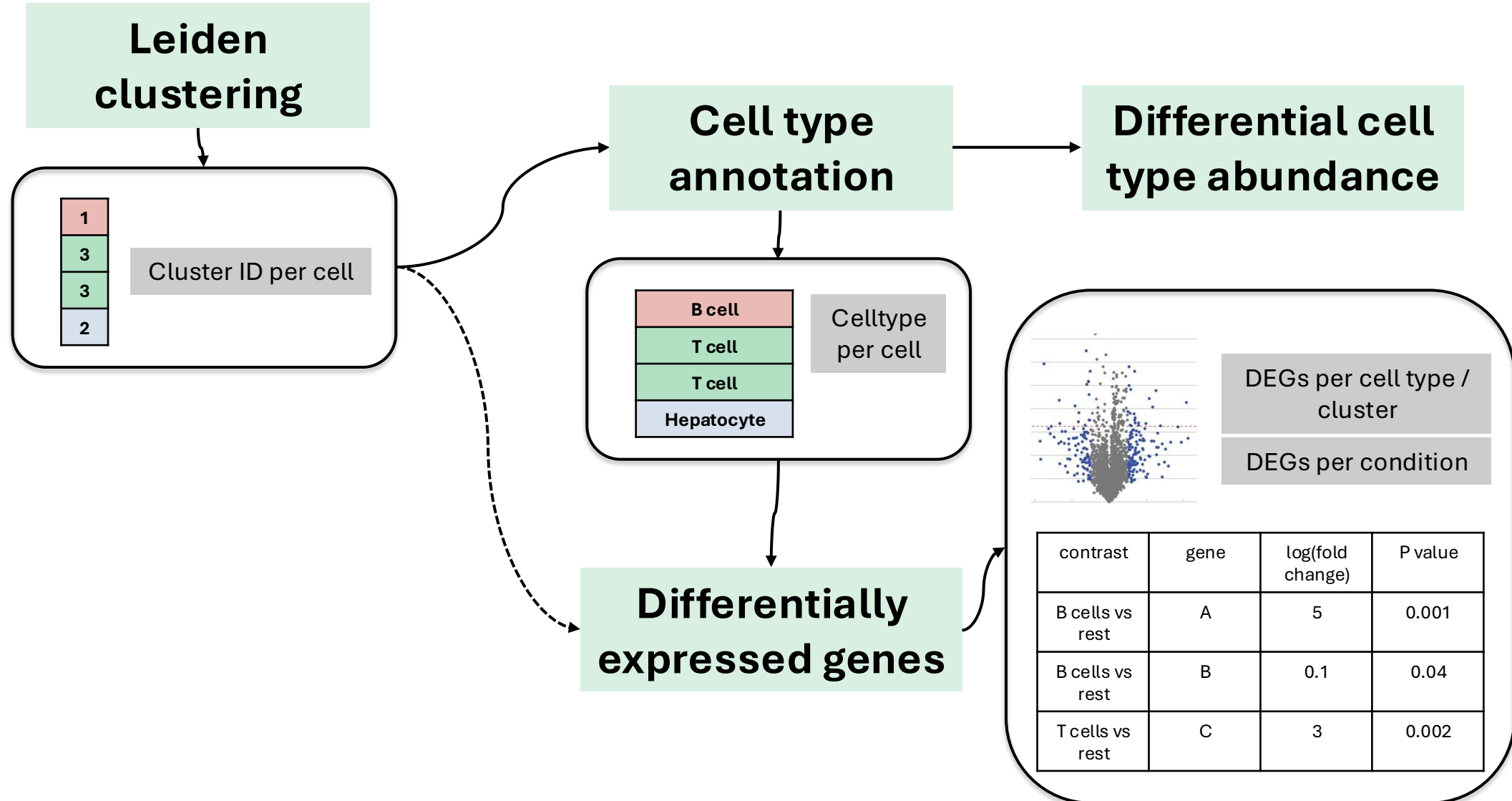
Turning clusters into knowledge with annotation and differential expression analysis



Turning clusters into knowledge with annotation and differential expression analysis



Turning clusters into knowledge with annotation and differential expression analysis



Cell type annotation $\approx$ giving names to clusters



What is a cell type?

Cell type annotation $\approx$ giving names to clusters



What is a cell type?

A cellular phenotype that is

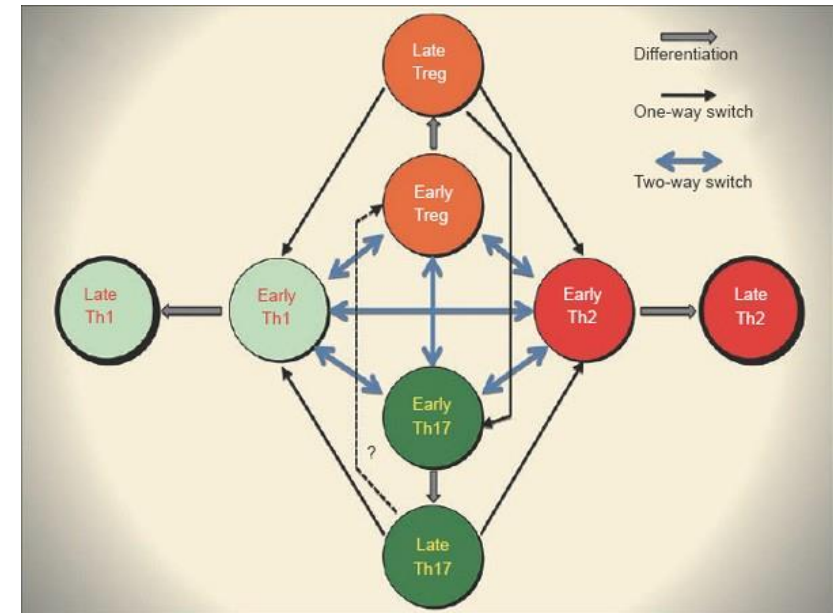
- robust across datasets
- Linked to specific functions
- Identifiable based on specific markers (most often proteins or transcripts)

Cell type annotation $\approx$ giving names to clusters



What is a cell type?

- A cellular phenotype that is
- robust across datasets
 - Linked to specific functions
 - Identifiable based on specific markers (most often proteins or transcripts)



cell state

cell type

Cell type annotation $\approx$ giving names to clusters



Why do we need to give names to clusters?



...could we not analyse everything
based on our computed embeddings,
trajectories and clusters?



Cell type annotation $\approx$ giving names to clusters

Why do we need to give names to clusters?

To speak about biology
with others in
established categories

To link sc omics based
results to decades of
immunological research





Cell type annotation $\approx$ giving names to clusters

Why do we need to give names to clusters?

To speak about biology
with others in
established categories

To link sc omics based
results to decades of
immunological research



The “pure ML” view: We should not let ourselves be biased by old papers based on few markers, instead reanalyse biology from scratch

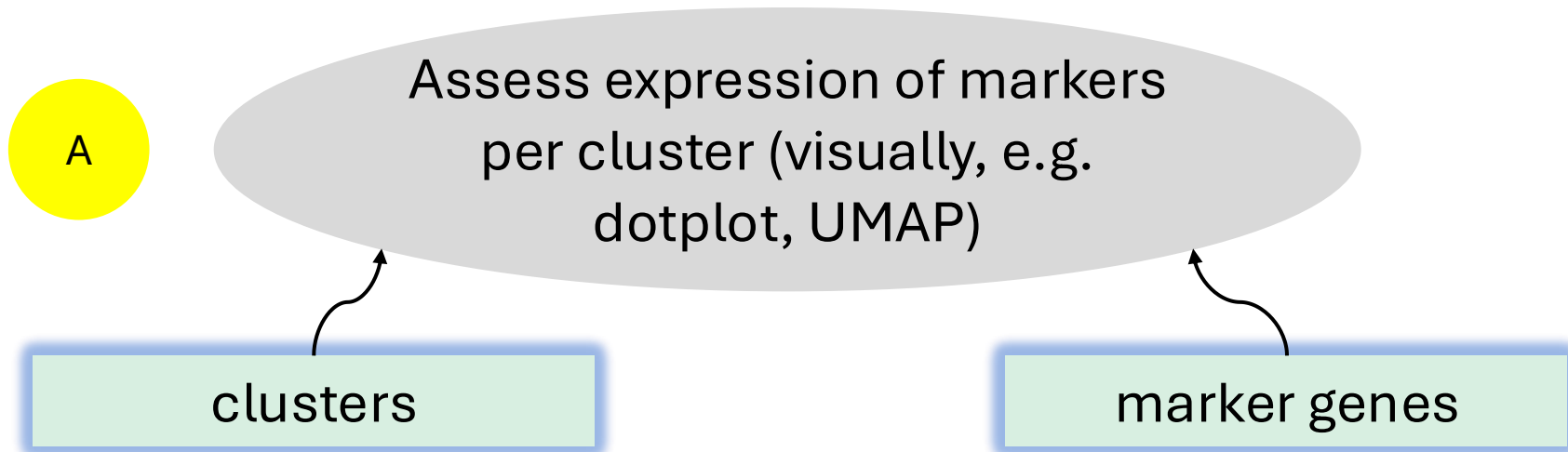
Manual cell type annotation



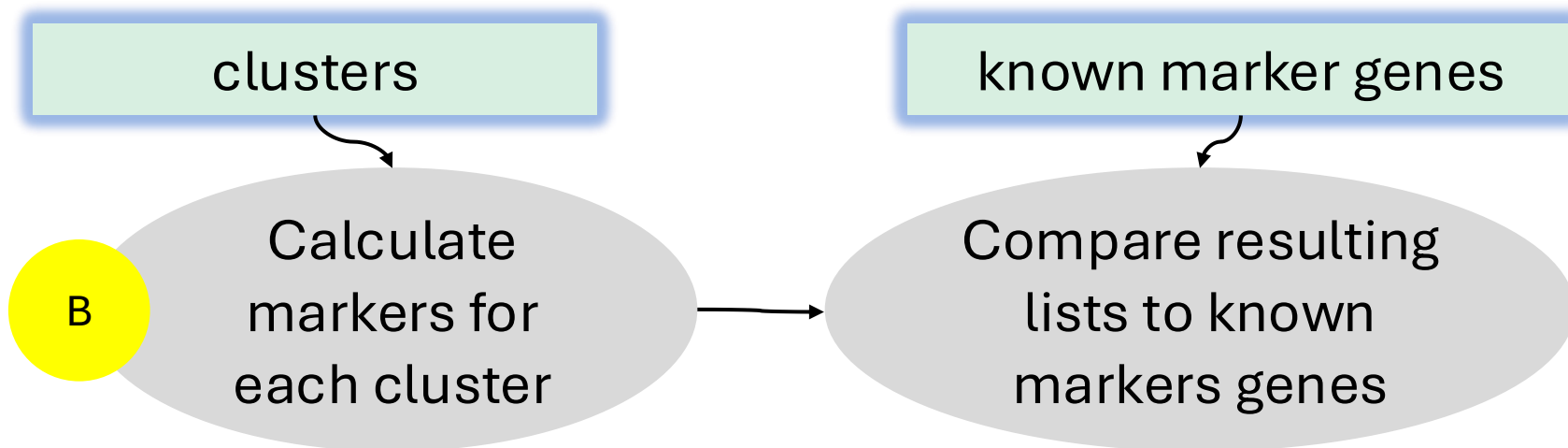
clusters

marker genes

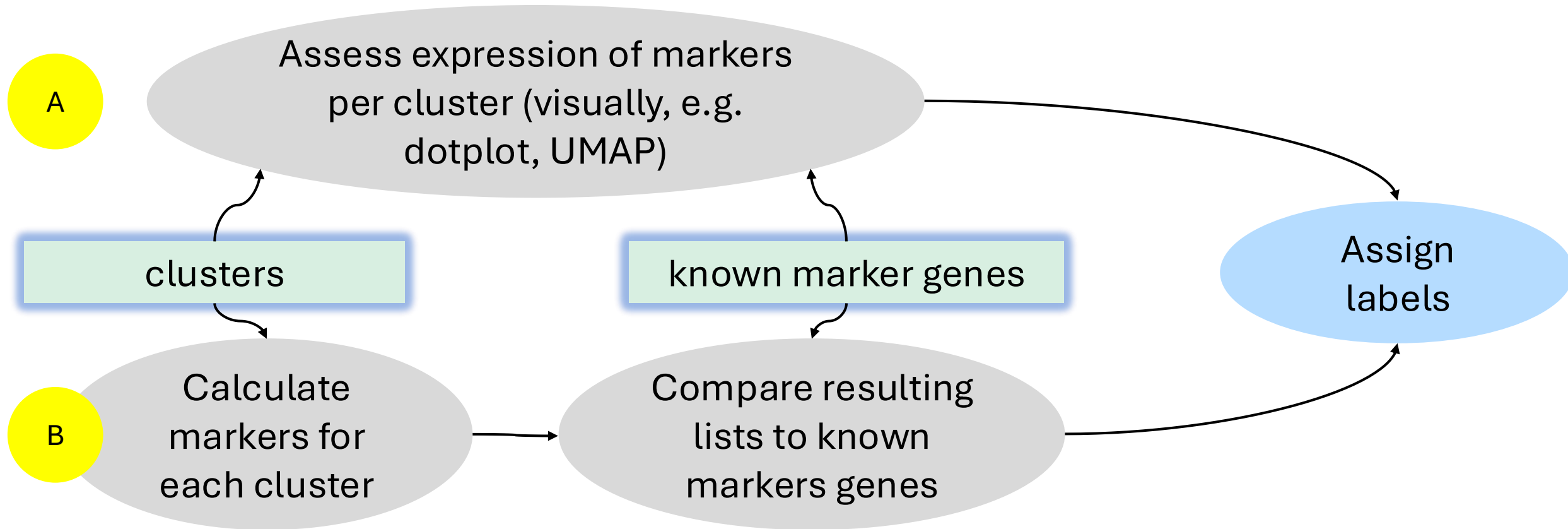
Manual cell type annotation



Manual cell type annotation



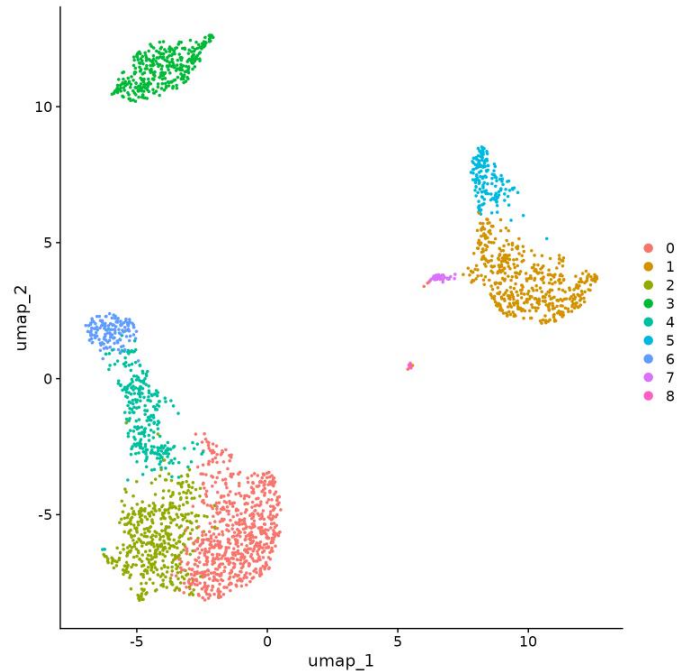
Manual cell type annotation



Method A - PBMC annotation / ingredients



clusters



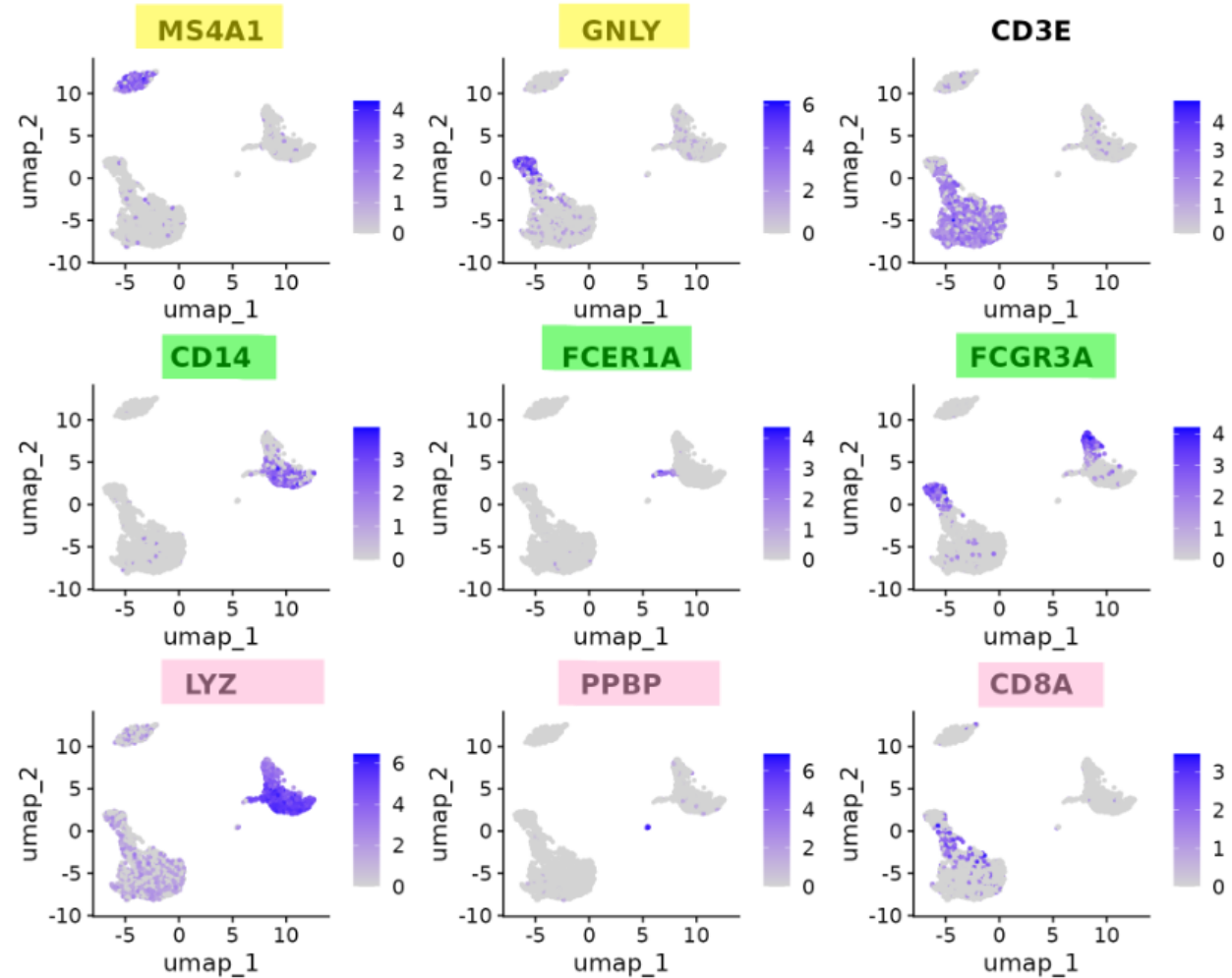
known marker genes

Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
1	CD14, LYZ	CD14+ Mono
2	IL7R, S100A4	Memory CD4+
3	MS4A1	B
4	CD8A	CD8+ T
5	FCGR3A, MS4A7	FCGR3A+ Mono
6	GNLY, NKG7	NK
7	FCER1A, CST3	DC
8	PPBP	Platelet

Method A - PBMC annotation / UMAP inspection



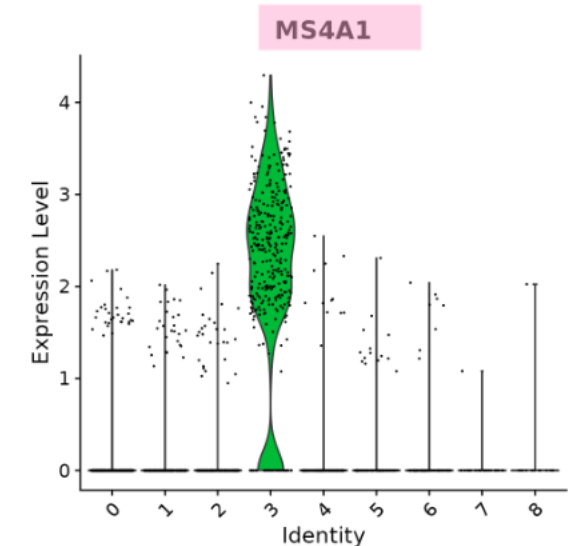
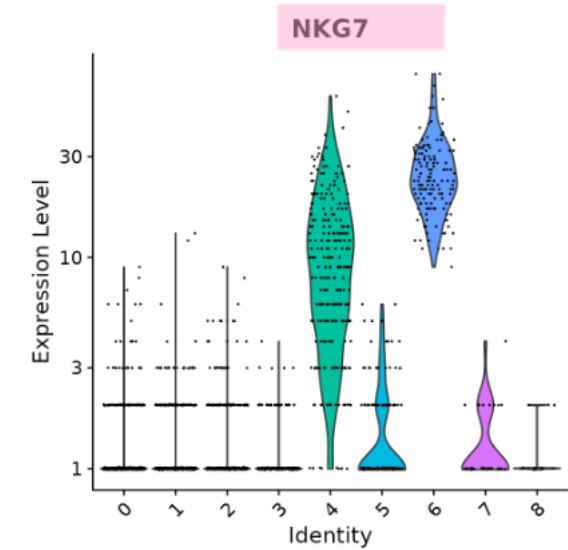
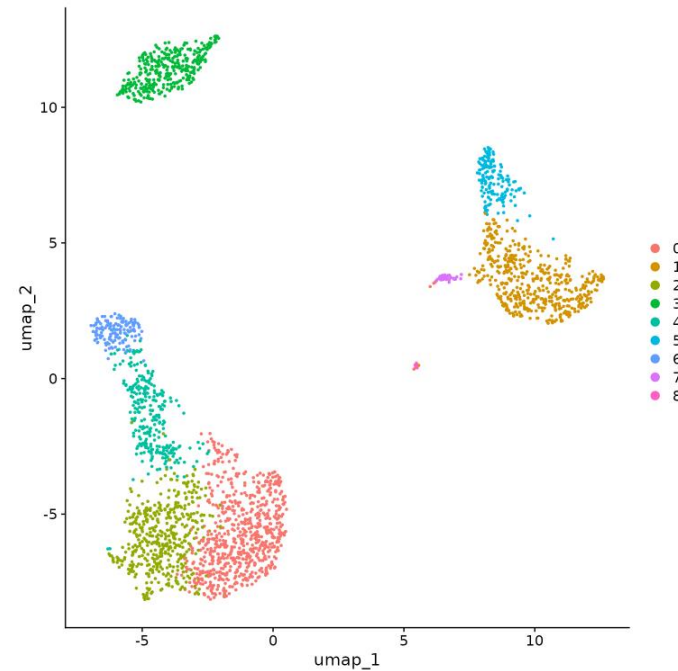
Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
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8	PPBP	Platelet



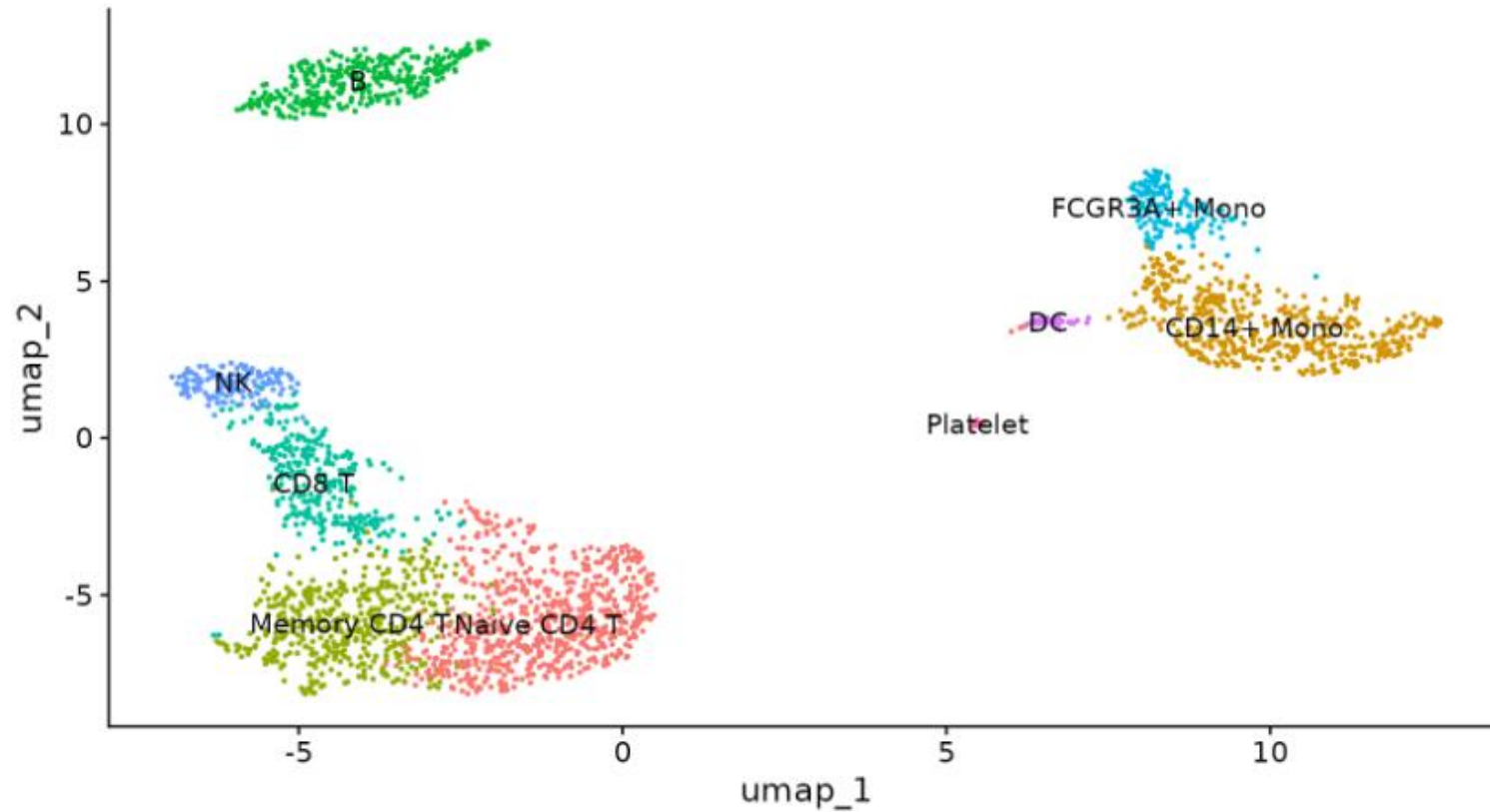
Method A - PBMC annotation / violin plots of marker genes



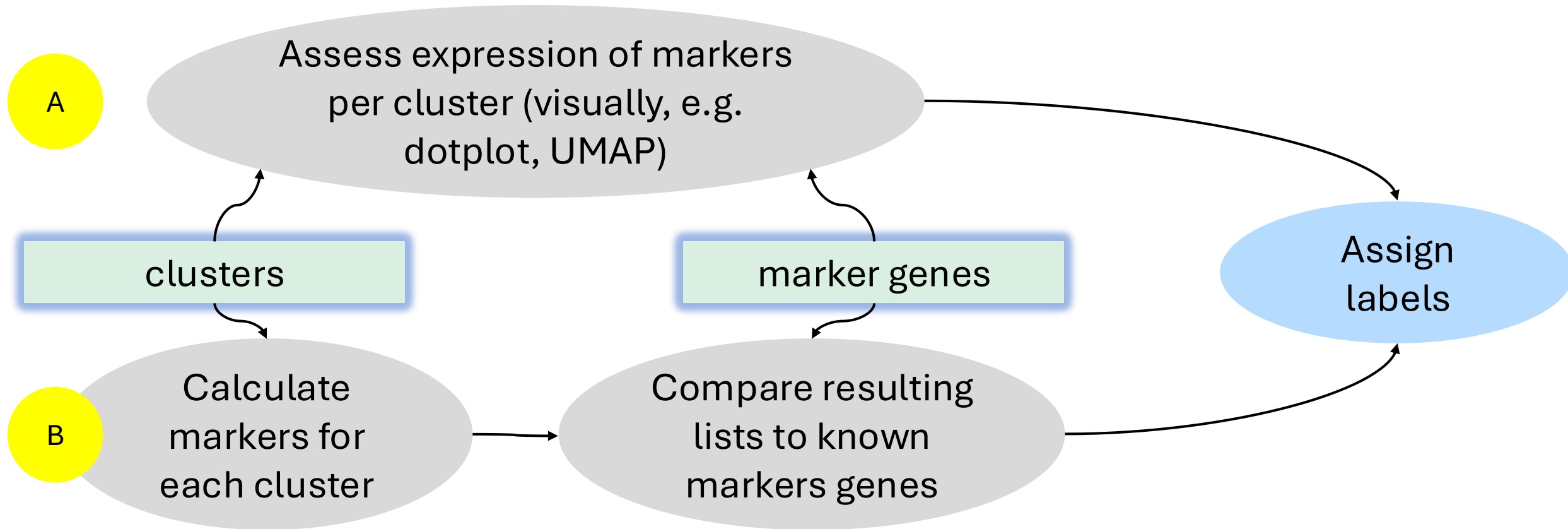
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0	IL7R, CCR7	Naive CD4+ T
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8	PPBP	Platelet



Method A - PBMC annotation / results



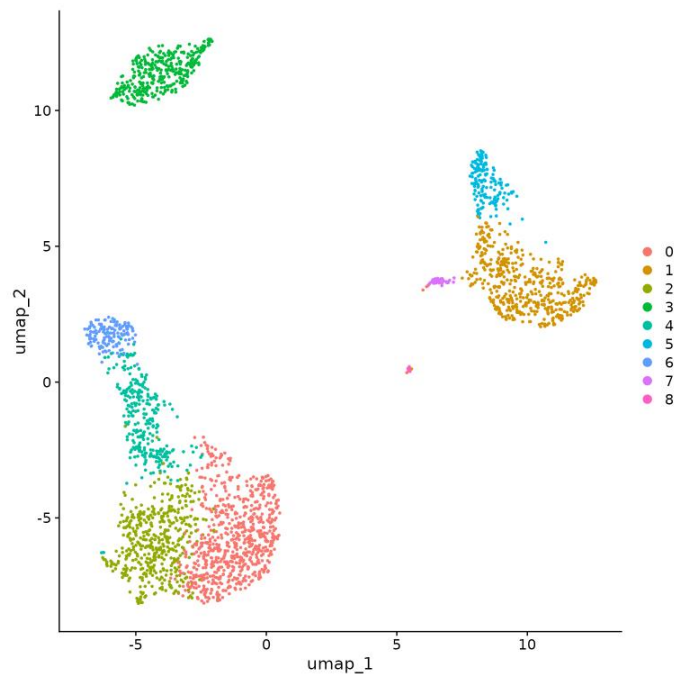
Manual cell type annotation



Method B - PBMC annotation / ingredients



clusters

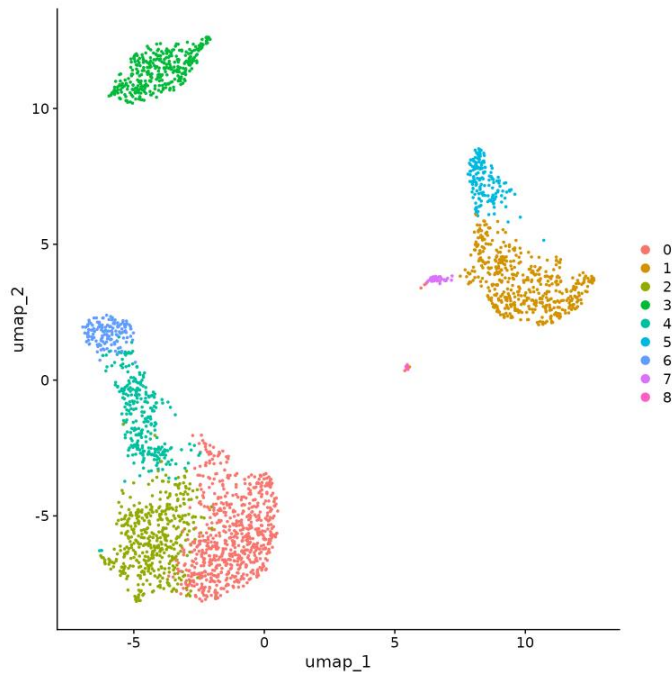


Method B - PBMC annotation / ingredients



clusters

Calculate
markers for
each cluster



Seurat

```
cluster0.markers <- FindMarkers(pbmc, ident.1 = 0, logfc.threshold = 0.25, test.use = "roc", only.pos = TRUE)
```

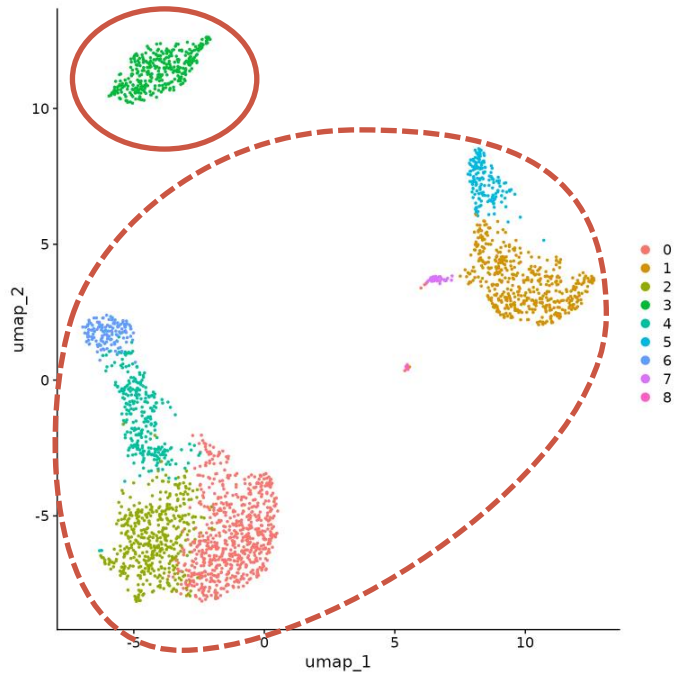
scanpy

```
sc.tl.rank_genes_groups(  
    adata, groupby="leiden_2", method="wilcoxon", key_added="dea_leiden_2"  
)
```

https://satijalab.org/seurat/articles/pbmc3k_tutorial

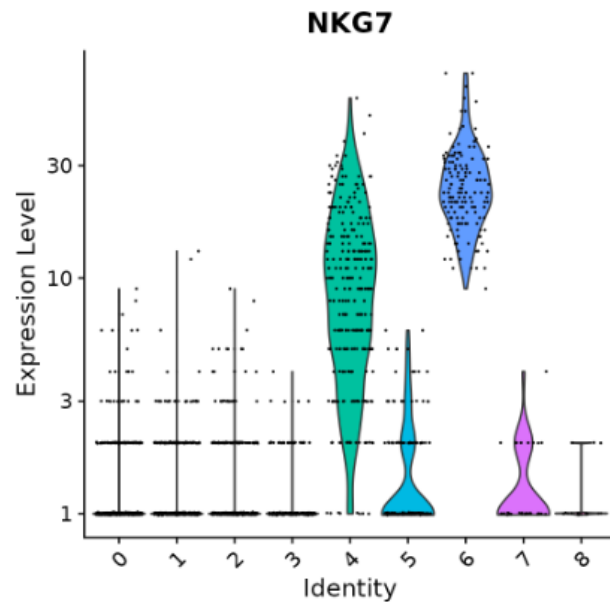
https://www.sc-best-practices.org/cellular_structure/annotation.html#manual-annotation

Calculating marker genes per cluster



1. Select cluster for comparison
(for marker gene detection we typically compare
“cluster X” vs “all the rest”)

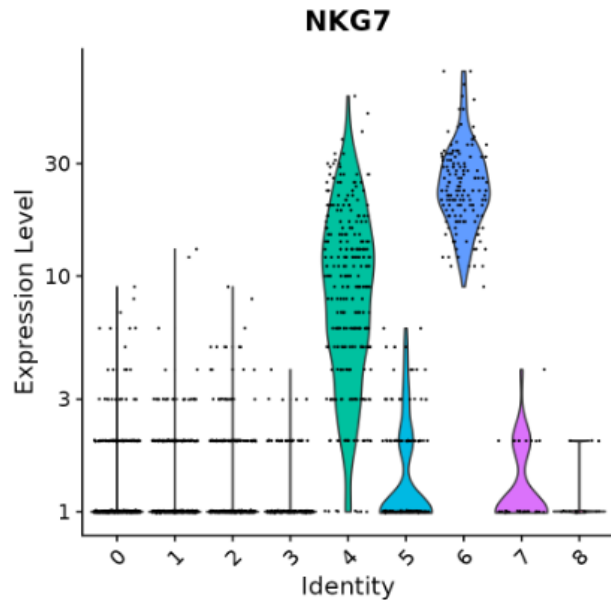
Calculating marker genes per cluster



1. Select cluster for comparison (X)

2. For each gene, perform statistical test that compares the expression distribution between X and rest

Calculating marker genes per cluster



1. Select cluster for comparison (X)

2. For each gene, perform statistical test that compares the expression distribution between X and rest

Classical testing

t-test

Wilcoxon's test

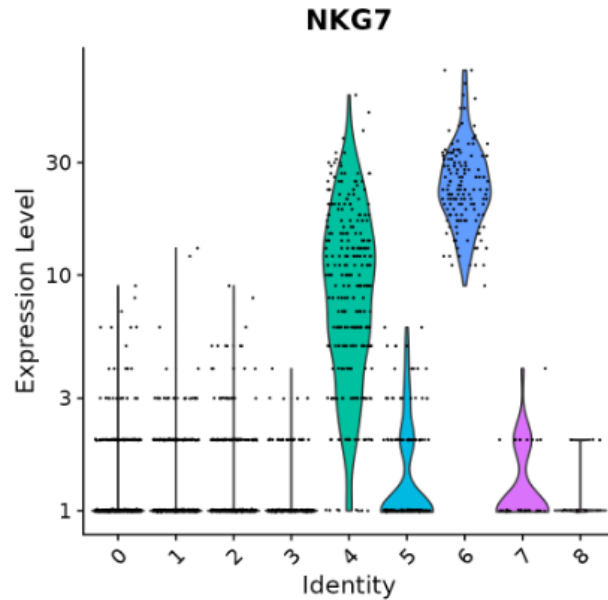
Classifier-based

Logistic
regression

ROC analysis

...

Calculating marker genes per cluster



1. Select cluster for comparison (X)

2. For each gene, perform statistical test that compares the expression distribution between X and rest

3. Apply multiple testing correction to resulting p-values

Calculating marker genes per cluster



p_val	avg_logFC	pct.1	pct.2	p_val_adj	cluster	gene
1.783378e-11	0.6126976	0.327	0.441	2.966472e-07	0	CAP1
1.111275e-05	0.5448506	0.201	0.277	1.848495e-01	0	NDUFA2
5.572391e-69	0.9152596	0.705	0.325	9.269115e-65	1	IL7R
1.834643e-44	0.7485499	0.308	0.086	3.051746e-40	1	MAL
0.000000e+00	4.4769283	0.968	0.207	0.000000e+00	2	S100A9
0.000000e+00	4.1251610	0.932	0.114	0.000000e+00	2	S100A8
1.124982e-304	4.1155861	0.992	0.192	1.871295e-300	3	NKG7
9.974215e-141	4.0851525	0.596	0.113	1.659111e-136	3	GNLY
0.000000e+00	2.7777043	0.934	0.043	0.000000e+00	4	CD79A
1.689832e-184	2.3696654	1.000	0.818	2.810866e-180	4	CD74
6.129817e-167	2.2891146	0.954	0.138	1.019634e-162	5	FCGR3A
1.497900e-118	2.3893744	1.000	0.316	2.491607e-114	5	LST1
8.448866e-05	0.6377758	1.000	0.956	1.000000e+00	6	RPL36
1.450393e-03	0.5964958	0.459	0.288	1.000000e+00	6	GYPC
2.002848e-181	7.1596296	1.000	0.011	3.331538e-177	7	PF4
6.022917e-101	8.5050281	1.000	0.025	1.001852e-96	7	PPBP

1. Select cluster for comparison (X)

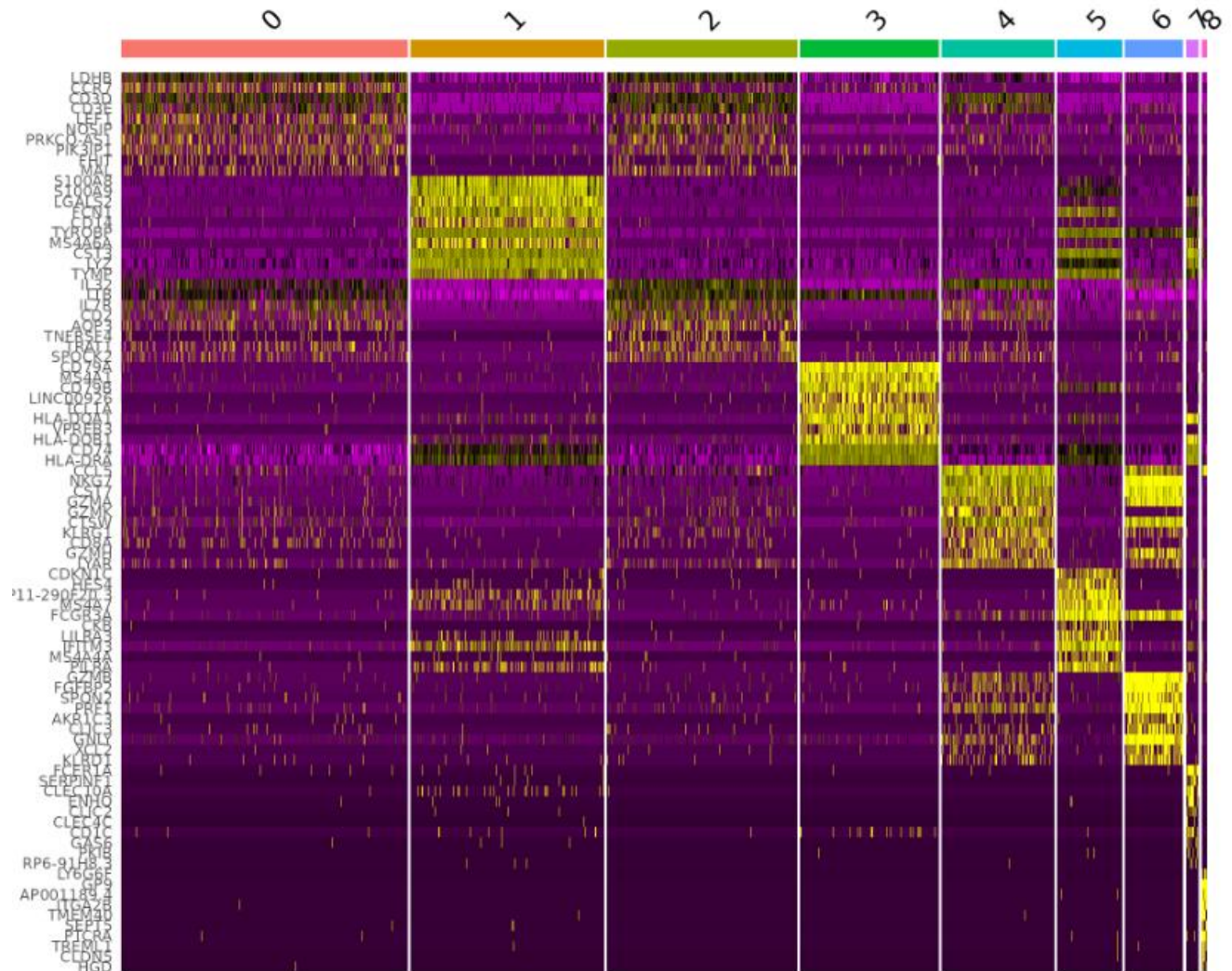
2. For each gene, perform statistical test that compares the expression distribution between X and rest

3. Apply multiple testing correction to resulting p-values

4. Identify significant / most important marker genes by sorting + thresholding **fold changes** and **p-values**



clusters



https://satijalab.org/seurat/articles/pbm3k_tutorial

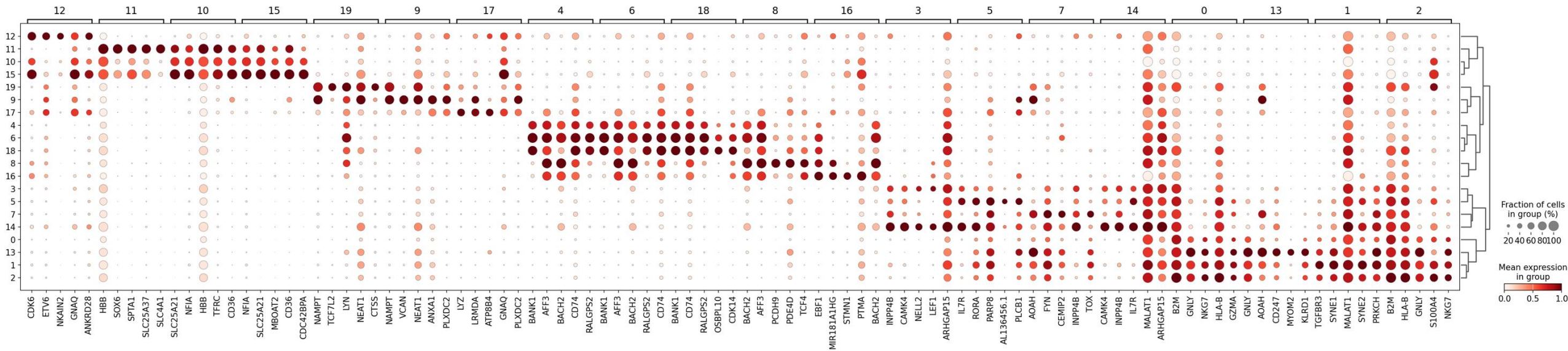
<https://www.sc-best->

practices.org/cellular_structure/annotation.html#manual-annotation

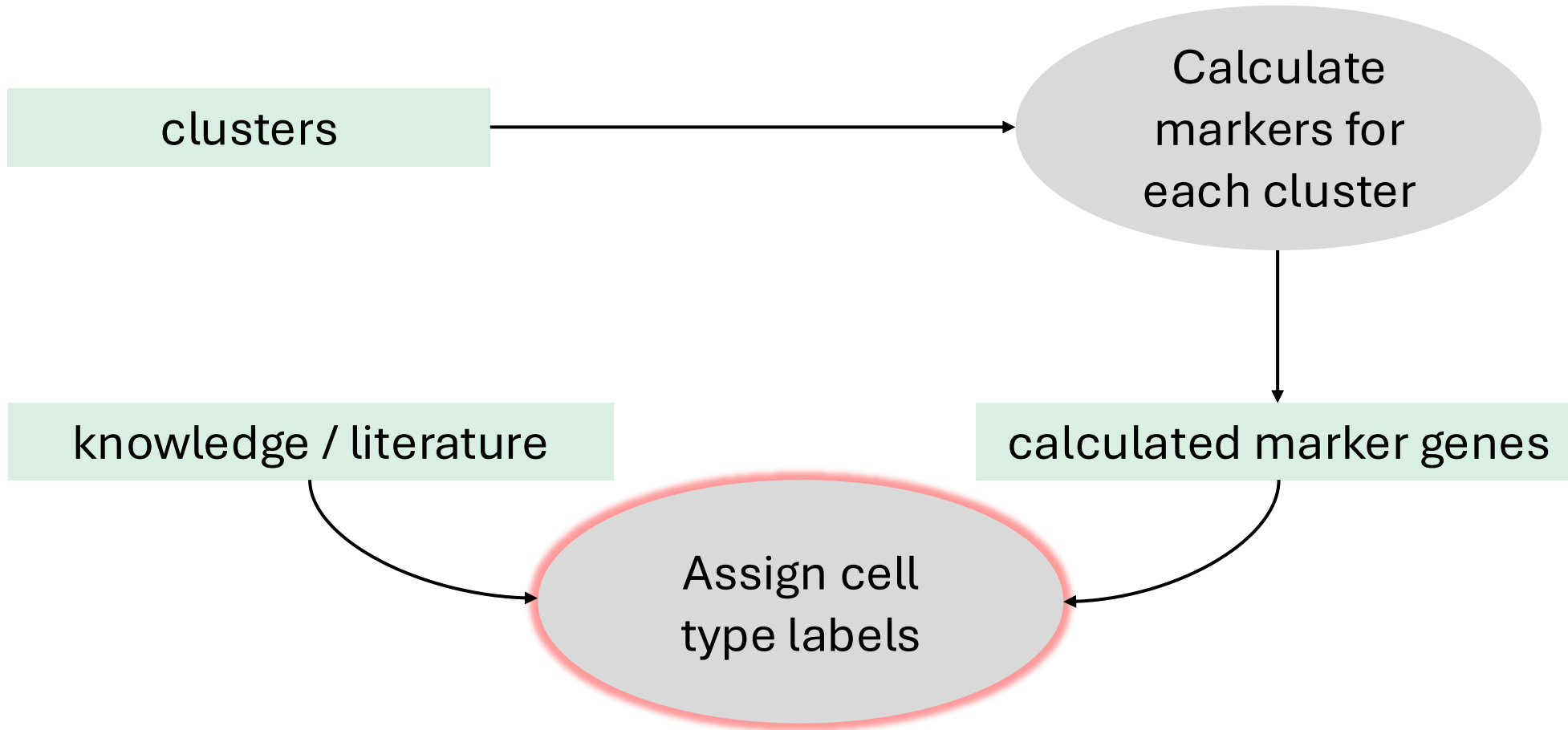
Method B - PBMC annotation / marker dotplot



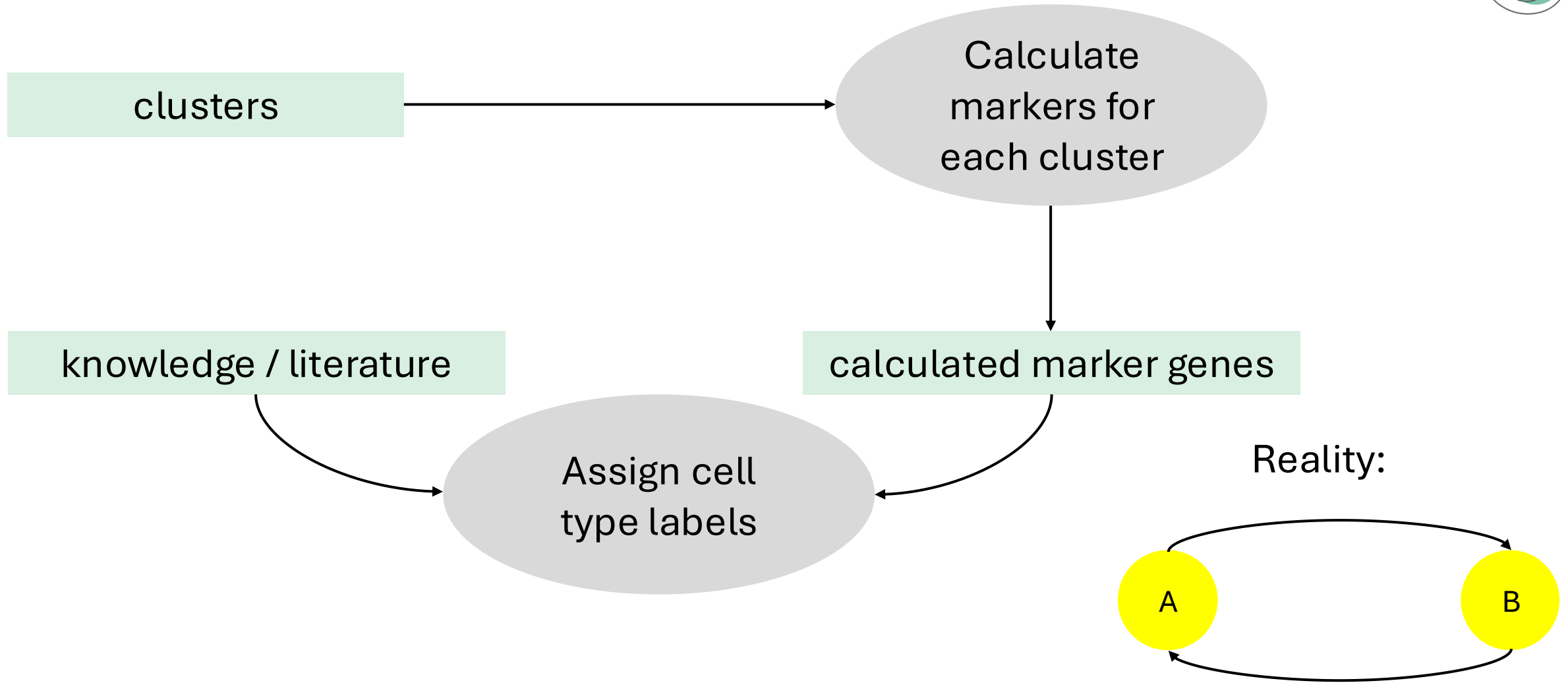
calculated marker genes



Method B - PBMC annotation / ingredients



Method B - PBMC annotation / ingredients



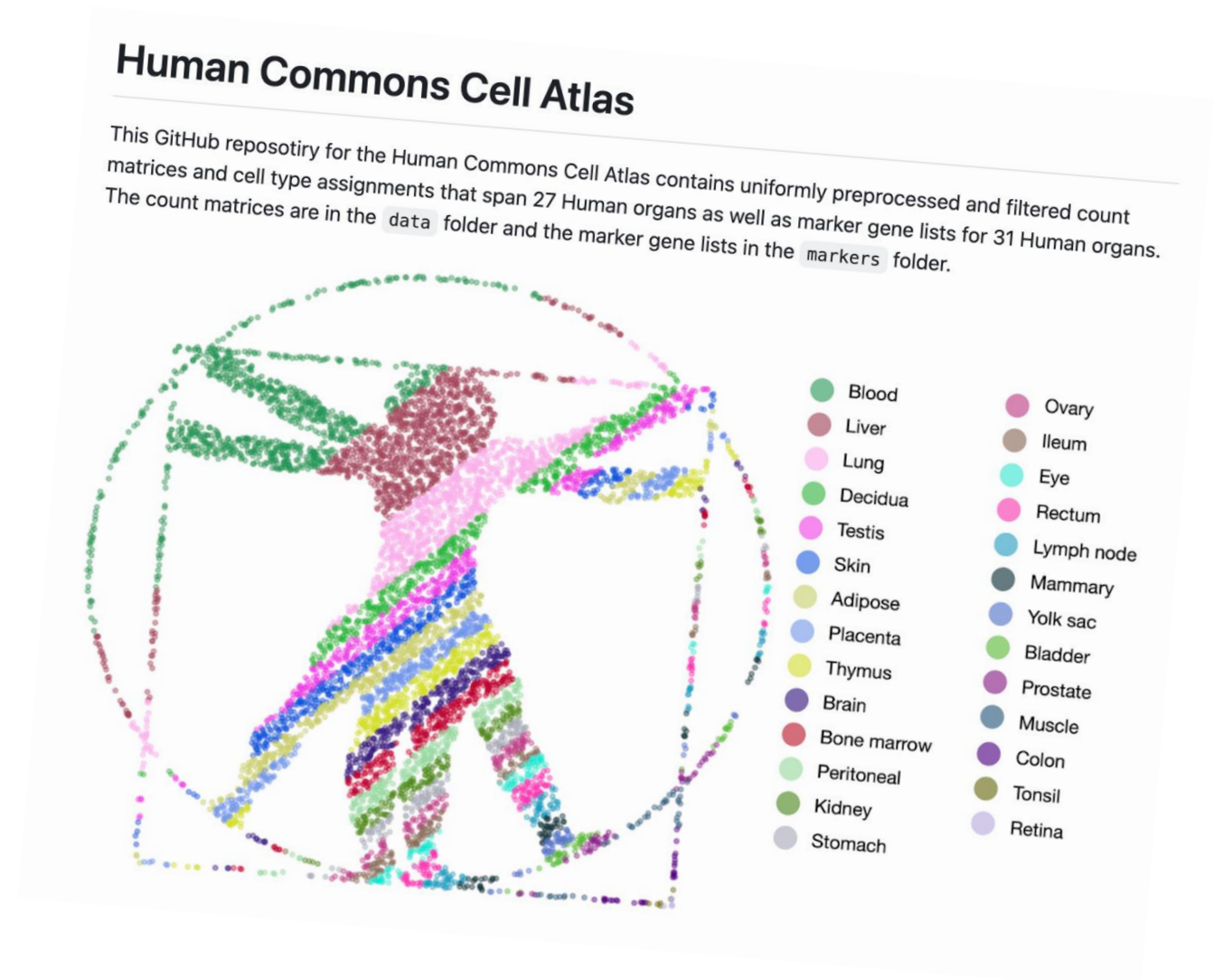
Where to find marker genes?



General literature

Annotated single-cell
datasets

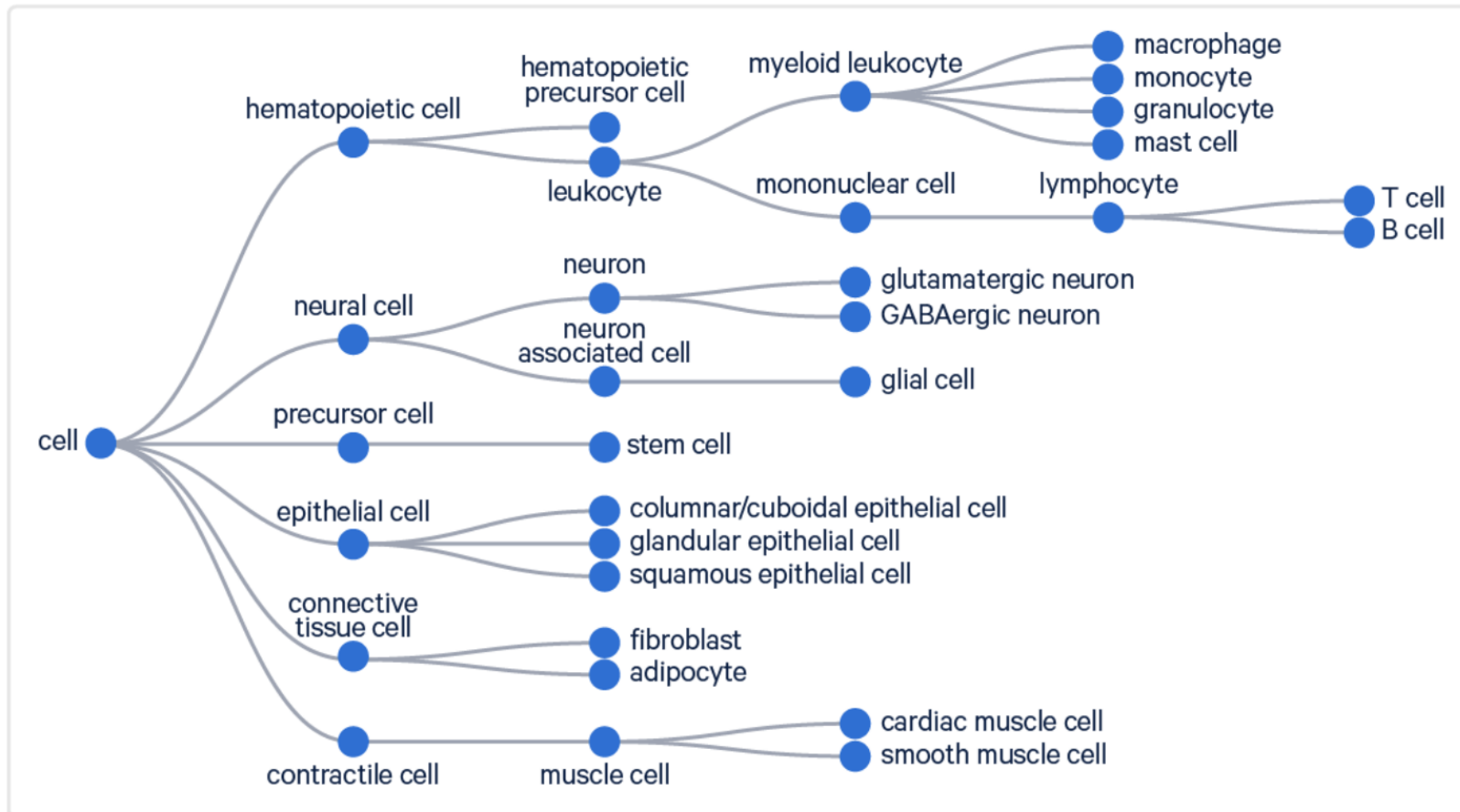
Dedicated marker gene
resources



Unifying cell type names: the “Cell Ontology” project



Cell Ontology aims to classify, structure and provide uniform labels for cell types across organisms (<https://obofoundry.org/ontology/cl.html>, <https://www.ebi.ac.uk/ols4/ontologies/cl>)



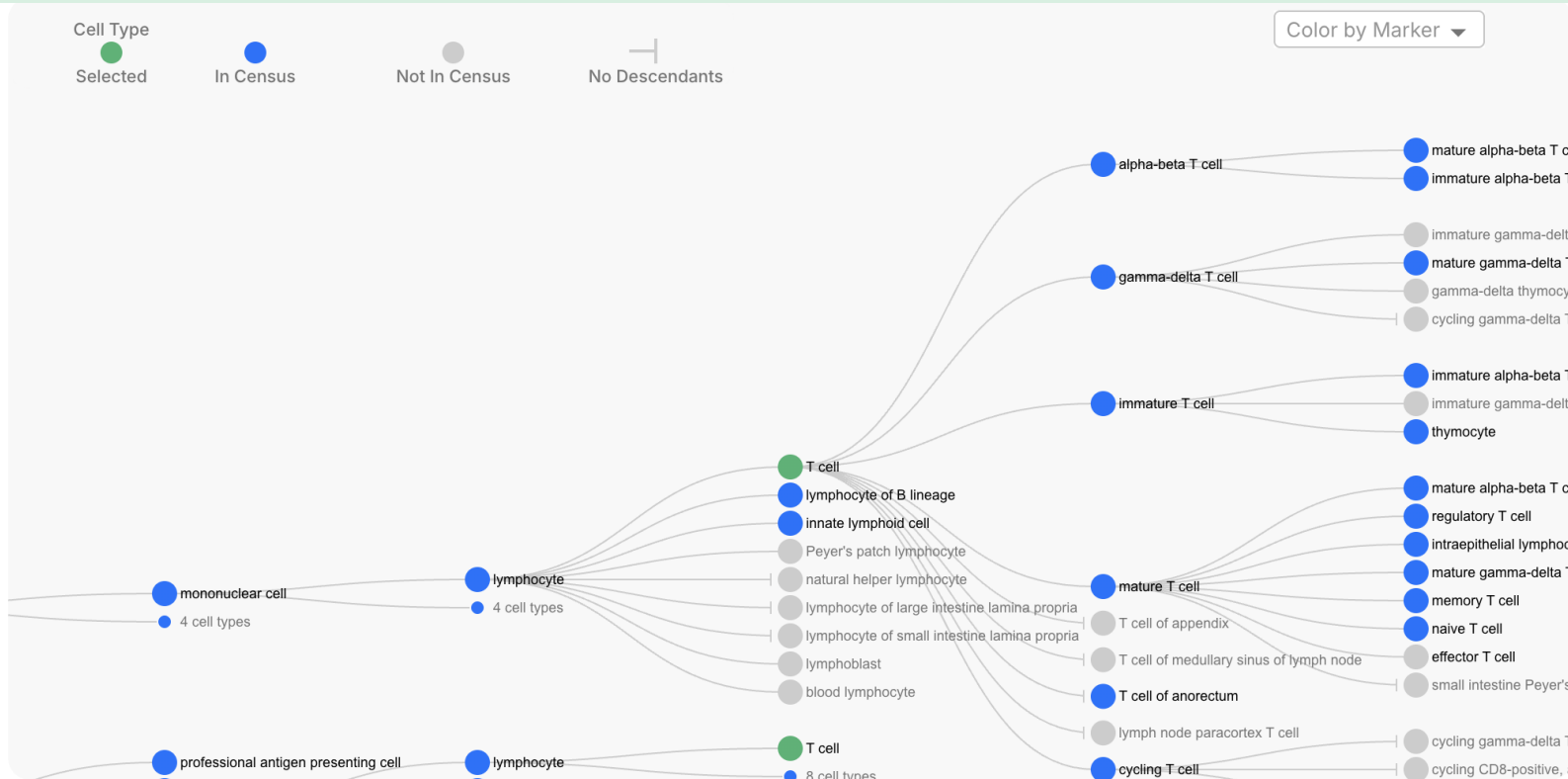
CELLXGENE project:

<https://cellxgene.cziscience.com/>

Unifying cell type names: the “Cell Ontology” project



Cell Ontology aims to classify, structure and provide uniform labels for cell types across organisms (<https://obofoundry.org/ontology/cl.html>, <https://www.ebi.ac.uk/ols4/ontologies/cl>)



The cellxgene CellGuide (<https://cellxgene.cziscience.com/cellguide>) provides ontological as well as marker gene information for many cell types

← example: T cell

Labelling cells the way others do is advised!

Automated annotation methods



!

Automated annotation methods currently serve as a starting point for annotation, not as an endpoint. Quality depends on:

Annotation
method

Quality of
training data

Automated annotation methods



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Automated annotation methods currently serve as a starting point for annotation, not as an endpoint. Quality depends on:

Annotation
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Quality of
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Similarity
between training
and query data

Automated annotation methods



!

Automated annotation methods currently serve as a starting point for annotation, not as an endpoint. Quality depends on:

Annotation
method

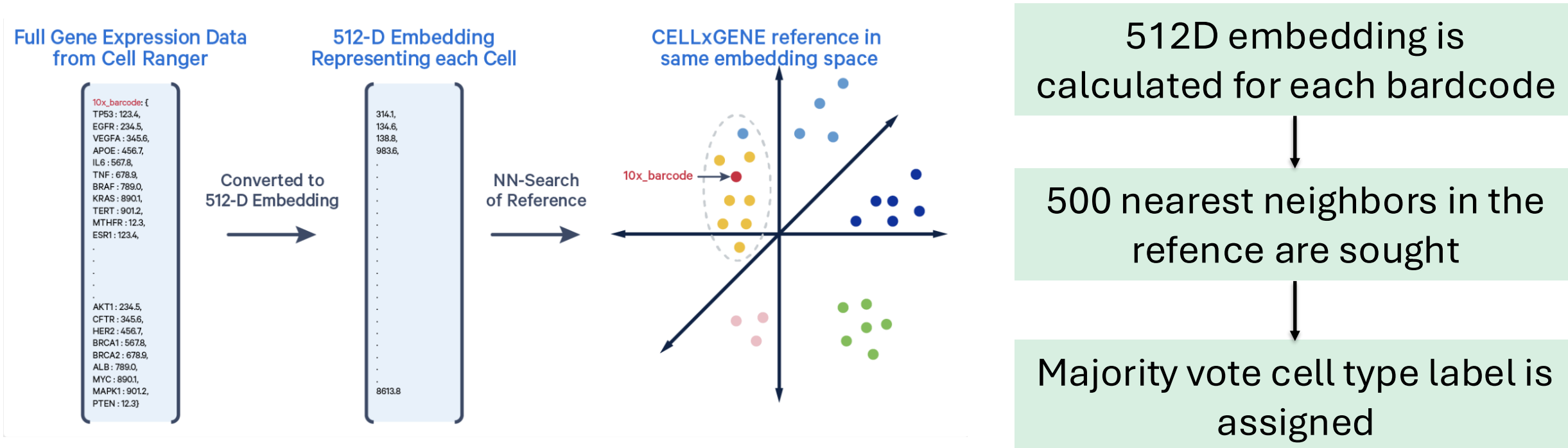
Quality of
training data

Similarity
between training
and query data

- Marker gene-based methods (e.g. Garnett, CellAssign)
- Classifier trained on previous datasets/atlas (e.g. **CellTypist**, Clustifyr, panAzimuth)
- Reference mapping (e.g. scArches, Symphony)

2026: Automated annotation is moving upstream in the workflow

cellranger (v9 and later) can include automated cell type annotation and v10.1 added support for the Pan-Human Azimuth model (runs locally).



Finally, an annotated dataset.

